Towards Efficient LLM Hallucination Detection Using Lightweight Models via Cross-Dataset Evaluation

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***Abstract*—Hallucination in large language models (LLMs) where generated responses appear fluent but contain factually incorrect information poses a significant challenge for reliable Artificial Intelligence (AI) deployment. This paper proposes a lightweight and explainable hallucination detection framework that extracts four compact features from question-answer pairs: semantic similarity, Natural Language Inference (NLI) score, entity overlap, and answer length. Six models are evaluated [LogestIC Regression (LR), Random Forest (RF) , XG-Boost,DistilBERT,TinyBERT and MiniLM] across three English benchmarks [HaluEval, TruthfulQA, and FActScore], using a cross-dataset generalization strategy. The best result is achieved by MiniLM with 66.66% accuracy, while Logistic Regression operates 100,000 times faster than GPT-4 at zero cost.SHapley Additive exPlanations(SHAP) analysis reveals Answer Length as the dominant feature, exposing a systematic bias.An Arabic extension using AraHalluEval achieves 50.66% accuracy. A temperature effect experiment further demonstrates that hallucination rates increase from 20% to 80% as generation temperature rises from 0.1 to 1.5. Results confirm that lightweight models provide an efficient and interpretable alternative to large evaluator models.**

I. Introduction

Large Language Models (LLMs) have achieved strong performance in natural language processing(NLP) by generating fluent and contextually relevant text. However, they still suffer from hallucination, where the generated output may be factually incorrect, unsupported, or logically inconsistent. This problem is especially critical in high risks domains such as medicine, education, and software engineering, where inaccurate responses can lead to harmful consequences [1], [8], [11].

Hallucinations occur partly because LLMs are mainly trained to predict the next token rather than explicitly verify factual correctness. As a result, they may produce convincing answers that are not grounded in reliable results. Existing hallucination detection methods include consistency checking, retrieval-based verification, Natural Language Inference (NLI), and large model-based evaluation. Although these methods are effective, many of them depend on expensive evaluator models, external retrieval systems, or repeated sampling, which limits their efficiency and practical deployment [2], [4], [16], [19], [21], [25].

To address this gap, this study proposes a lightweight and interpretable hallucination detection framework. Instead of building a new LLM or relying on costly verification pipelines, the proposed framework extracts four compact features from each [question , answer] pair: semantic similarity, NLI score, entity overlap, and answer length. These features are used to train and evaluate six models: Logistic Regression, Random Forest, XGBoost, DistilBERT, TinyBERT, and MiniLM.

The framework is evaluated on three English hallucination related benchmarks: HaluEval, TruthfulQA, and FActScore. A cross-dataset evaluation strategy is used, where models are trained on one dataset and tested on unseen datasets to measure generalization. The best English cross-dataset result was achieved by MiniLM, with 66.66% accuracy when trained on TruthfulQA and tested on FActScore. In addition, an Arabic extension using AraHalluEval is included to examine the applicability of the framework to Arabic hallucination detection, where the best result reached 50.66% accuracy.

The main contribution of this work is an efficient, reproducible, and explainable hallucination detection pipeline that reduces computational cost while maintaining interpretability. The study also highlights the difficulty of cross-dataset and multilingual hallucination detection, showing that hallucination patterns vary across datasets, languages, and model types.

II. Background

Large Language Models (LLMs) are transformer-based neural models trained on large-scale textual data to generate context-aware responses. By predicting the next token in a sequence, these models can produce fluent and coherent text across a wide range of natural language processing tasks. However, fluency does not guarantee factual correctness, and LLMs may generate answers that appear convincing while containing inaccurate or unsupported information [1], [6].

A major limitation of LLMs is hallucination, which occurs when generated content is factually incorrect, logically inconsistent, or unsupported by the input or external knowledge [1]. This issue is partly caused by the probabilistic nature of language modeling, where outputs are generated based on learned statistical patterns rather than explicit factual verification or grounded reasoning [22]. As a result, hallucinated responses may be difficult to detect because they often appear linguistically fluent and confident.

Several approaches have been proposed to detect hallucination. Consistency-based methods examine whether a model produces stable answers under repeated or reformulated prompts. Retrieval-based methods compare generated content with external knowledge sources. Natural Language Inference (NLI) methods assess whether a generated statement is entailed, contradicted, or unsupported by a given input [2], [4]. In addition, semantic similarity and entity overlap can be used to measure whether the generated answer is aligned with the input in meaning and factual entities.

These approaches provide complementary signals for hallucination detection. However, many existing methods depend on large evaluator models, retrieval systems, or repeated sampling, which increase computational costs. Therefore, lightweight hallucination detection has become an important research direction. In this work, four compact features are used to represent each question–answer pair: semantic similarity, NLI score, entity overlap, and answer length. These features provide an interpretable and efficient basis for training models to classify answers as hallucinated or non-hallucinated.

III. Literature Review

Although Large Language Models (LLMs) are powerful, they often "hallucinate" by generating facts that are incorrect or inconsistent. This lack of reliability limits their use in practical settings [1], [7]. To solve this, researchers are developing ways to detect and prevent these errors, using everything from mathematical theory to complex system upgrades [6], [15]. This review organizes these efforts and identifies the future of the field. Table I presents a summary of the reviewed studies.

A. Fundamentals of Hallucination in Large Language Models

Early research established the conceptual foundations of hallucination in natural language generation, defining it as content that is inconsistent with the source or unsupported by it. This led to the distinction between intrinsic hallucinations, which contradict the input, and extrinsic hallucinations, which introduce unverifiable information [1]. These initial studies attributed hallucinations to both data-related issues and model-level factors, including exposure bias and decoding strategies, while also highlighting the limitations of traditional evaluation metrics.

Further analysis emphasized that hallucinations are inherently linked to the probabilistic nature of transformer-based architectures, where outputs are generated through next-token prediction rather than grounded reasoning [22]. Subsequent work introduced a broader classifications framework, categorizing hallucinations into representation-level, reasoning-level, and output-level errors, and distinguishing hallucination from factuality as a separate concept related to real-world consistency [5], [6].

B. Hallucination Detection Methods

Research on hallucination detection initially focused on consistency-based approaches, where model outputs are evaluated under repeated or reformulated queries. Multi-perspective consistency checking extended this idea by analyzing both response variability and query reconstruction, enabling the detection of overconfident yet incorrect outputs [2]. These methods provided a foundation for black-box detection without relying on external knowledge sources.

Later work incorporated more structured and robust techniques. Metamorphic testing introduced controlled input transformations to expose logical inconsistencies, while paraphrase-based constraints ensured output stability under semantic variations [4], [16]. In parallel, discriminative models were developed to explicitly classify hallucinated outputs, demonstrating improved reliability across different datasets and model architectures [20].

C. Fine-Grained and Entity-Level Detection

As detection techniques evolved, research shifted toward fine-grained analysis of hallucinations. Early approaches demonstrated that hallucinations can occur at the token level, motivating the development of decoding strategies that improve alignment between generated text and source information [12]. These methods enabled more precise identification of ungrounded content within generated sequences.

More advanced approaches introduced entity-level verification, where named entities and their relationships are extracted and validated against input data or external knowledge [13]. Recent developments further extended this idea through dynamic retrieval and hierarchical verification, allowing real-time detection and multi-granularity analysis of hallucinations at both sentence and entity levels [19], [25].

D. Evaluation Metrics and Benchmarking

The evaluation of hallucination has evolved alongside detection methods. Early work highlighted the limitations of traditional metrics, which often fail to capture factual correctness despite high linguistic quality [1]. This led to the development of evaluation frameworks that distinguish between hallucination detection and factuality assessment, providing a more comprehensive understanding of model reliability.

Recent studies have focused on faithfulness metrics, which measure how closely generated content aligns with source information. Approaches such as LLM-as-a-judge have shown strong correlation with human evaluation; however, they introduce potential bias due to shared model characteristics [15]. Additionally, structured benchmarks and controlled datasets have enabled more rigorous and reproducible evaluation of hallucination detection systems [3], [10].

E. Mitigation Strategies and Optimization

Beyond detection, significant research has been directed toward mitigating hallucinations. Retrieval-based approaches validate generated content against external knowledge sources,

TABLE I
GROUPED COMPARISON OF HALLUCINATION DETECTION METHODS IN LLMS

Category	Method / Model	Dataset	Metrics	Key Contribution	Limitations
Consistency-Based Methods [2], [4], [16], [19]	Self-consistency, mutation, and paraphrasing.	TruthfulQA, WikiBio, and QA datasets.	Accuracy, F1-score, AUC.	Detect hallucination through internal consistency without relying on external knowledge.	High computational cost due to repeated sampling, paraphrasing, or query reformulation.
Retrieval-Based Methods [10], [18], [26]	RAG, post-hoc retrieval, and hybrid verification.	Multi-domain QA and real-world datasets.	Accuracy, ROUGE, BERTScore.	Improve factual accuracy by grounding generated content in external knowledge sources.	Performance depends on retrieval quality and usually increases latency.
NLI / Semantic-Based Methods [13], [20], [21], [25]	Entailment models, semantic classification, and entity matching.	HaluEval, FEVER, and FactCC.	Precision, Recall, F1-score.	Provide interpretable hallucination detection using semantic and contradiction signals.	Sensitive to NLI model accuracy, dataset imbalance, and domain differences.
Training-Based Methods [12], [23], [9]	Controlled decoding, model modification, and fine-tuning.	WikiBio and MSCOCO.	BLEU, Accuracy, F1-score.	Reduce hallucination during generation rather than only detecting it after generation.	Require retraining, model access, or higher implementation complexity.
Domain-Specific Methods [8]	Static and dynamic verification for code.	Code datasets.	Precision, Recall, F1-score.	Achieve strong performance in specialized hallucination detection scenarios.	Limited generalization outside the target domain.
Analytical Models [17]	Mathematical modeling of hallucination probability.	HaluEval.	Error rate.	Quantify hallucination likelihood and analyze factors affecting generation reliability.	Limited direct applicability in real-world detection pipelines.
Arabic Hallucination Evaluation [30]	Fine-grained hallucination evaluation framework for Arabic LLMs.	AraHalluEval.	Accuracy, hallucination rate, and category-level analysis.	Introduces an Arabic hallucination benchmark with fine-grained hallucination categories across Arabic and multilingual LLMs.	Focuses mainly on evaluation; further work is needed on lightweight cross-dataset detection and multilingual transfer.
Survey and Review Studies [1], [3], [5], [6], [7], [11], [14], [15], [22], [24]	Taxonomies, evaluation frameworks, and system-level analysis.	Multiple datasets.	Various metrics.	Provide comprehensive understanding of hallucination definitions, causes, evaluation, and mitigation strategies.	No direct implementation or experimental validation.

while hybrid frameworks combine multiple strategies to improve reliability across different tasks [10]. In structured domains such as software engineering, hybrid methods integrating static and dynamic analysis have proven effective in identifying and correcting hallucinated outputs [8].

More recent work has explored optimization-based strategies, including the impact of decoding parameters on hallucination behavior. Studies have shown that parameters such as temperature, top-k, and top-p influence the randomness of generation, where higher values increase hallucination risk and controlled tuning improves reliability [17]. Hierarchical semantic frameworks further enhance mitigation by enabling multi-level correction of generated content [25]. Recent scholarship examines hallucination mitigation in industrial contexts, highlighting the tension between systemic reliability and operational efficiency [26].

F. Emerging Trends and Applications

Recent research has expanded into new directions, including multimodal hallucination and application-specific challenges. Multimodal models introduce additional complexity by requiring consistency between textual and visual inputs, leading to new types of hallucination that require specialized detection techniques [14]. At the same time, efficiency has become a key concern, with lightweight and decoupled architectures demonstrating that effective detection can be achieved with reduced computational cost [21].

In addition to technical advancements, studies have begun exploring the societal implications of hallucinations. In educational contexts, hallucinated outputs can lead to misinformation and reduced critical thinking, highlighting the importance of AI literacy and responsible usage [11]. Novel approaches, including semantic and affective alignment strategies, further illustrate the ongoing shift toward more robust and human-centered solutions [23].

G. Synthesis and Research Gaps

Despite significant progress, several critical challenges remain. Current approaches are predominantly reactive, lacking fully integrated architectures capable of preventing hallucinations during generation. There is also a persistent trade-off between efficiency and performance, particularly between lightweight models and computationally intensive evaluation frameworks [15], [21].

Moreover, while textual hallucination detection is well studied, multimodal and cross-domain scenarios remain underexplored. Finally, the long-term impact of hallucinations on user trust and decision-making, especially in high-stakes domains, continues to be insufficiently addressed, highlighting the need for more comprehensive and human-centered research directions [11], [18].

IV. METHODOLOGY

This section describes the methodology used to develop the proposed lightweight hallucination detection framework. The framework classifies generated answers as hallucinated or non-hallucinated by extracting compact and interpretable features from (question,answer) pairs. As shown in Figure 1, the pipeline includes dataset preparation, feature extraction, model training, cross-dataset evaluation, Arabic extension and result analysis.

A. Dataset Selection and Preprocessing

Four publicly available hallucination datasets were selected to cover both English and Arabic languages. Each dataset was

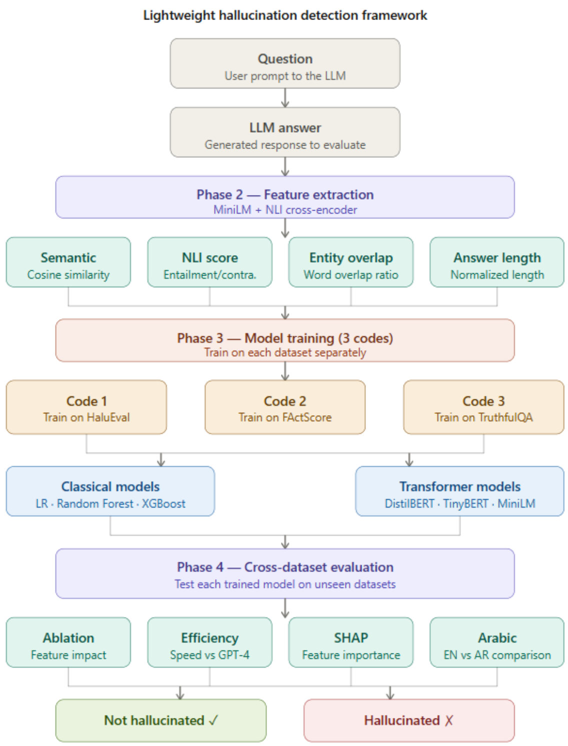


Fig. 1. Overview of the proposed lightweight hallucination detection framework.

preprocessed into a unified (question, answer, label) format, where label 0 indicates a non-hallucinated answer and label 1 indicates a hallucinated answer.

1) Dataset Selection Criteria: The selection of datasets for this study was guided by four primary criteria. **First**, each dataset must provide explicit binary hallucination annotations to support supervised classification. **Second**, the selected datasets should exhibit diversity in question format, answer style, and knowledge domain, beside enabling a comprehensive cross-dataset generalization evaluation.**Third**, to extend the framework beyond English and assess cross-lingual transferability, at least one Arabic hallucination benchmark was required. **Fourth**, all datasets must be publicly available to ensure full reproducibility of the experimental results. Based on these criteria, three English benchmarks were selected [HaluEval, TruthfulQA, and FActScore] also one Arabic benchmark[AraHalluEval], resulting in a total of 33,793 samples spanning both English and Arabic."

2) Datasets Description:

a) HaluEval: is an English hallucination evaluation benchmark . It contains 10,000 question-answer pairs sampled from a larger collection of LLM-generated responses. Each sample consists of a question, a supporting knowledge passage, and an answer labeled as hallucinated or non-hallucinated. The dataset covers a wide range of factual topics and is nearly balanced, with a hallucination rate of 50.1%, making it suitable as a primary training source[27].

b) TruthfulQA: is an English benchmark to evaluate the truthfulness of LLM-generated responses. It consists of 817 questions and contains 38 categories, including misconceptions, fiction, and conspiracy theories. Each question is accompanied by a set of correct and incorrect answers.In this study, correct answers are assigned label 0, and incorrect answers are assigned label 1, resulting in 5,918 samples with a hallucination rate of 56.1% [28].

c) FActScore: is an English factual precision benchmark . It evaluates the factual accuracy of LLM-generated biographical text by decomposing outputs into atomic facts and verifying each against a knowledge source. In this study, supported facts (label S) are assigned label 0 and unsupported facts (label NS) are assigned label 1, while irrelevant facts (label IR) are discarded. The resulting dataset contains 14,525 samples with a hallucination rate of 30.4% [29].

d) AraHalluEval: is an Arabic hallucination evaluation benchmark . It contains 4,200 question-answer pairs generated by 14 Arabic and multilingual LLMs, Labeled with fine-grained hallucination labels across eight categories, including Named-Entity Hallucination, Temporal/Number Hallucination, and Factual Contradiction. For this study, a binary label is derived: a sample is considered hallucinated if any hallucination type is present. After removing mixed-language and excessively long samples, 3,350 clean Arabic samples were retained with a hallucination rate of 49.4% [30]

3) Dataset Statistics: As shown in Table II and figure 2, the selected datasets vary in language, size, label distribution, and hallucination rate. HaluEval is nearly balanced, TruthfulQA contains more hallucinated samples, FActScore contains more non-hallucinated samples, and AraHalluEval provides the Arabic extension used in this study.

TABLE II
DATASET STATISTICS USED IN THIS STUDY

Dataset	Language	Samples	Hallucinated	Not Hallucinated	Hallucination Rate
HaluEval	English	10,000	5,010	4,990	50.1%
TruthfulQA	English	5,918	3,318	2,600	56.1%
FActScore	English	14,525	4,419	10,106	30.4%
AraHalluEval	Arabic	3,350	1,657	1,693	49.4%

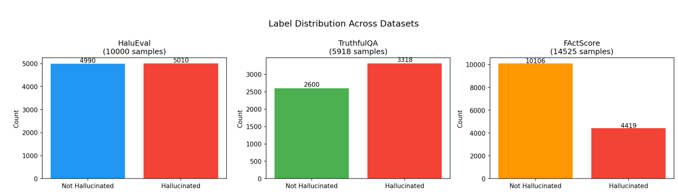


Fig. 2. Label distribution across the English hallucination detection datasets.

4) Preprocessing Steps: All datasets were preprocessed into a unified (question, answer, label) format before feature extraction. The following steps were applied: **Text cleaning**:

whitespace normalization and removal of special characters were applied to all samples. Arabic Unicode characters were preserved in AraHalluEval. **Label conversion**:hallucination labels were converted to binary values: 1 for hallucinated and 0 for non-hallucinated.

TruthfulQA:correct answers were assigned label 0 and incorrect answers were assigned label 1, resulting in multiple samples per original question.

FActScore:supported atomic facts (S) were assigned label 0, unsupported facts (NS) were assigned label 1, and irrelevant facts (IR) were discarded.

AraHalluEval :samples containing non-Arabic characters or exceeding 500 characters in length were removed to ensure data quality, reducing the dataset from 4,200 to 3,350 samples.

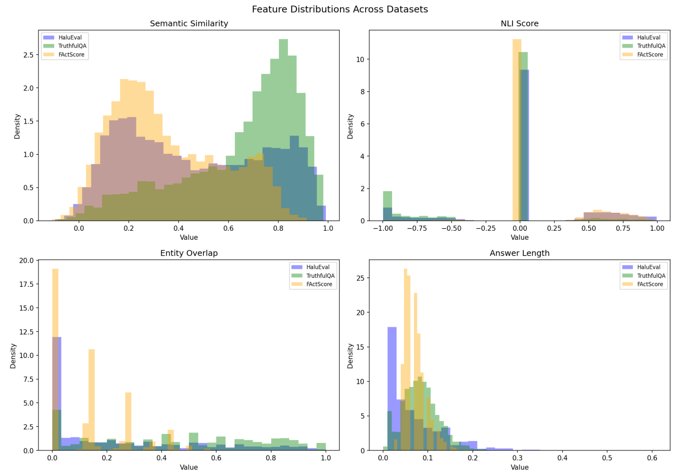


Fig. 3. Distribution of the extracted features across HaluEval, TruthfulQA, and FActScore.

B. Model Selection and Configuration

This section describes the models selected for hallucination detection. The framework utilizes two categories of models:**classical machine learning models and lightweight transformer-based models**. All models take as input a four-dimensional feature vector extracted from each question-answer pair and output a binary classification label.

1) Feature Extraction: Four lightweight features are extracted from each (question, answer) pair to represent the relationship between the question and the generated answer. These features are designed to capture semantic, syntactic, and structural properties of the answer without relying on external knowledge sources.

a) Semantic Similarity: measures the cosine similarity between the sentence embeddings of the question and the answer using the MiniLM model (sentence-transformers/all-MiniLM-L6-v2). A lower similarity score may indicate that the answer is factually inaccurate in relation to the question.

b) NLI Score: measures the degree of entailment or contradiction between the question and the answer using a Natural Language Inference cross-encoder (cross-encoder/nli-MiniLM2-L6-H768). Positive scores indicate entailment, and negative scores indicate contradiction.

c) Entity Overlap: computes the ratio of shared words between the question and the answer. A low overlap may suggest that the answer introduces entities not present in the question.

d) Answer Length: measures the normalized length of the answer, computed as the number of words divided by 100, bounded at 1.0.This feature captures the tendency of hallucinated answers to be longer or shorter than factual ones.

As shown in Figure 3, the three English datasets show different patterns in semantic similarity, NLI score, entity overlap, and answer length. These differences indicate that hallucination characteristics vary across datasets, which supports the need for cross-dataset evaluation.Table III shows the lightweight feature extraction for models.

TABLE III
FEATURE DESCRIPTION

Feature	Model Used	Description
Semantic Similarity	MiniLM	Cosine similarity between the question and answer.
NLI Score	NLI cross-encoder	Entailment or contradiction score between the question and answer.
Entity Overlap	Rule-based	Word overlap ratio between the question and answer.
Answer Length	Rule-based	Normalized answer length.

2) Classical Models: Three classical machine learning models were selected for their simplicity, interpretability, and computational efficiency. All three models are trained on the four-dimensional feature vector, and all of these models have a fixed random seed of 42.

a) Logistic Regression: a linear classifier that models the probability of hallucination as a function of the input features. It was configured with a maximum of 1,000 iterations.

b) Random Forest: an ensemble of decision trees that aggregates predictions through majority voting. It was configured with 100 estimators, a maximum depth of 10 to prevent overfitting.

c) XGBoost: a gradient boosting classifier that builds an ensemble of weak learners sequentially. It was configured with 100 estimators, log loss as the evaluation metric.

As shown in Table IV, the classical models were configured using standard hyperparameters suitable for lightweight feature-based classification.

TABLE IV
MODEL CONFIGURATION FOR CLASSICAL MODELS

Model	Key Hyperparameters
Logistic Regression	`max_iter=1000`
Random Forest	`n_estimators=100`, `max_depth=10`
XGBoost	`n_estimators=100`, `eval_metric=logloss`

3) Transformer Models: Three lightweight transformer-based models were selected for their balance between performance and computational efficiency. All three models were fine-tuned for binary sequence classification using the [question , answer] pair as input, formatted as "question [SEP] answer", with a maximum sequence length of 512 tokens, a learning rate of .00002, and 3 training epochs.

a) DistilBERT: a distilled version of BERT that retains 97% of BERT's language understanding while being 40% smaller and 60% faster. It contains 66 million parameters with a model size of 268MB.[31]

b) TinyBERT: a further compressed version of BERT obtained through knowledge distillation. It contains 14.5 million parameters with a model size of 62.7MB, making it the most lightweight transformer in this study. [32]

c) MiniLM: a compact transformer model that achieves strong performance through deep self-attention distillation. It contains 33 million parameters with a model size of 133MB. [33]

Table V shows the transformer models were configured with the same learning rate and number of epochs to ensure a fair comparison.

TABLE V
TRANSFORMER MODEL CONFIGURATION

Model	Size	Parameters	Learning Rate	Epochs
DistilBERT	268MB	66M	2e-5	3
TinyBERT	62.7MB	14.5M	2e-5	3
MiniLM	133MB	33M	2e-5	3

C. Evaluation Strategy

The evaluation strategy is designed to evaluate the cross-dataset generalization capability of the proposed framework. Rather than training and testing on the same dataset, **each model is trained on one dataset and evaluated on the remaining unseen datasets**, simulating a realistic deployment scenario where the target domain is unknown.

1) Cross-Dataset Evaluation: A leave-one-out cross-dataset evaluation strategy is applied, in which three experimental configurations are defined as follows:

a) Code 1 : Train HaluEval: Models are trained on HaluEval and evaluated on TruthfulQA and FActScore

b) Code 2 : Train FActScore: Models are trained on FActScore and evaluated on HaluEval and TruthfulQA.

c) Code 3 :Train TruthfulQA: Models are trained on TruthfulQA and evaluated on HaluEval and FActScore. This strategy ensures that no overlap exists between training and test data, providing an unbiased measure of cross-dataset generalization.

Table VI summarizes the Cross-Dataset Evaluation Strategy.

TABLE VI
CROSS-DATASET EVALUATION DESIGN

Code	Train Dataset	Test Datasets
Code 1	HaluEval	TruthfulQA + FActScore
Code 2	FActScore	HaluEval + TruthfulQA
Code 3	TruthfulQA	HaluEval + FActScore

2) Arabic Evaluation: To evaluate cross-lingual transferability, all classifiers trained on English datasets are evaluated directly on AraHalluEval without any retraining or fine-tuning. This zero-shot cross-lingual evaluation measures the ability of English-trained models to generalize to Arabic hallucination patterns.

3) Evaluation Metrics: The following metrics are used to evaluate model performance: **Accuracy , Precision , Recall , F1-Score , and ROC-AUC** . These metrics provide a comprehensive assessment of classification performance, especially in the presence of class imbalance.

V. RESULTS AND DISCUSSIONS

This section presents the experimental results of the proposed lightweight hallucination detection framework. The results are organized into eight subsections covering: **cross-dataset generalization, English vs. Arabic comparison, ablation study, efficiency analysis, explainability, error analysis, baseline comparison, and temperature effect analysis**.

A. Cross-Dataset Results

This section presents the cross-dataset evaluation results for all six models across the three experimental configurations. Results are reported separately for English and Arabic to facilitate a comprehensive assessment of both cross-dataset and cross-lingual generalization.

1) English Results: Table VII summarizes the cross-dataset accuracy results for all six models across the three experimental configurations . As shown, cross-dataset generalization remains challenging across all models and configurations. The best overall result is achieved by **MiniLM** with **66.66%** accuracy when trained on TruthfulQA and tested on FActScore (C3→FS). Among classical models, **Random Forest** achieves the highest accuracy of **58.44%** when trained on FActScore and tested on HaluEval (C2→HE). **Logistic Regression** demonstrates the most consistent performance across all configurations, suggesting that simpler models may generalize better than complex ones in cross-dataset settings. Figure 4 illustrates the results heatmap, while Figures 5–10 present the confusion matrices and ROC curves for each experimental code. **(C1=Code 1, C2=Code 2, C3=Code 3, TQA=TruthfulQA, FS=FActScore, HE=HaluEval)**

TABLE VII
CROSS-DATASET ACCURACY RESULTS (%)

Model	C1→TQA	C1→FS	C2→HE	C2→TQA	C3→HE	C3→FS
Logistic Regression	55.47	54.05	53.95	41.43	37.48	34.28
Random Forest	54.90	42.71	58.44	51.86	48.72	47.23
XGBoost	54.87	44.41	53.05	49.31	44.79	49.27
DistilBERT	44.73	42.90	48.50	47.89	43.48	65.65
TinyBERT	44.14	30.56	50.22	47.82	44.43	45.20
MiniLM	52.53	30.53	49.09	45.22	59.14	66.66

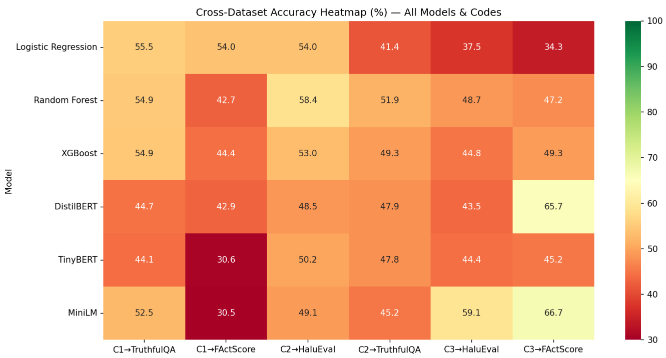
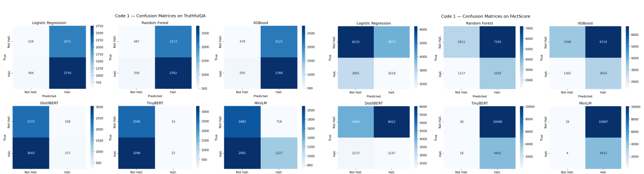


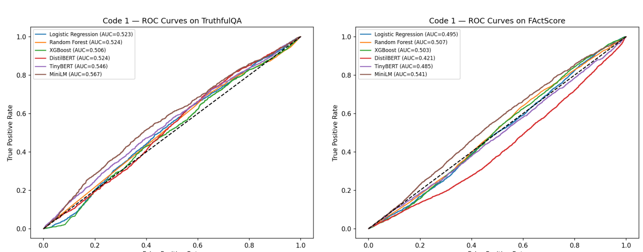
Fig. 4. Cross-dataset performance heatmap for all models and training settings.

Figure 5 and Figure 6 present the confusion matrices and ROC curves for Code 1, where the models were trained on HaluEval and tested on TruthfulQA and FActScore.



(a) Test on TruthfulQA (b) Test on FActScore

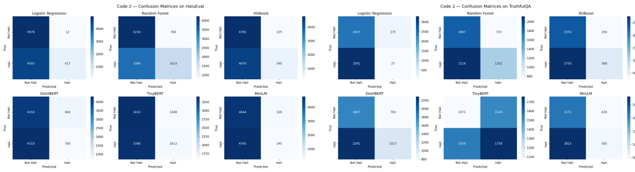
Fig. 5. Confusion matrices for Code 1 trained on HaluEval.



(a) Test on TruthfulQA (b) Test on FActScore

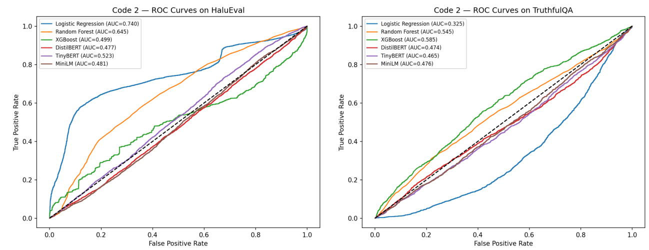
Fig. 6. ROC curves for Code 1 trained on HaluEval.

Figure 7 and Figure 8 present the confusion matrices and the ROC curves for Code 2, where the models were trained on FActScore and tested on HaluEval and TruthfulQA.



(a) Test on HaluEval (b) Test on TruthfulQA

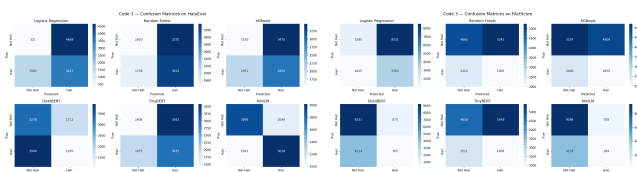
Fig. 7. Confusion matrices for Code 2 trained on FActScore.



(a) Test on HaluEval (b) Test on TruthfulQA

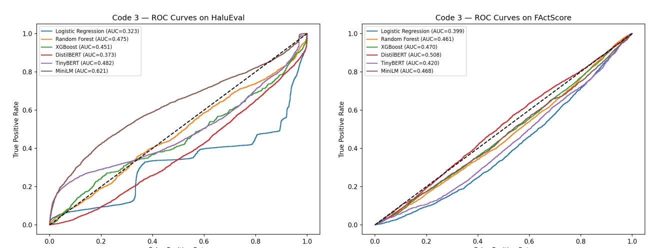
Fig. 8. ROC curves for Code 2 trained on FActScore.

Figure 9 and Figure 10 present the confusion matrices and the ROC curves for Code 3 where the models are trained on TruthfulQA and tested on HaluEval and FActScore.



(a) Test on HaluEval (b) Test on FActScore

Fig. 9. Confusion matrices for Code 3 trained on TruthfulQA.



(a) Test on HaluEval (b) Test on FActScore

Fig. 10. ROC curves for Code 3 trained on TruthfulQA.

2) Arabic Results: Table VIII shows the results reveal that models trained on FActScore generalize best to Arabic, achieving near **50%** accuracy across all three classical models. Models trained on HaluEval show the weakest Arabic performance, with accuracy dropping below 45% for all models. This suggests that HaluEval-specific features, particularly Answer Length bias, do not transfer well to Arabic text. Overall, all models perform close to random chance on Arabic, highlighting the challenge of zero-shot cross-lingual hallucination detection and the need for Arabic-specific training data.

TABLE VIII
ARABIC CROSS-DATASET ACCURACY RESULTS (%)

Code	Train Dataset	Model	Arabic Accuracy
Code 1	HaluEval	Logistic Regression	42.93
Code 1	HaluEval	Random Forest	44.93
Code 1	HaluEval	XGBoost	45.16
Code 2	FActScore	Logistic Regression	45.76
Code 2	FActScore	Random Forest	50.66
Code 2	FActScore	XGBoost	50.24
Code 3	TruthfulQA	Logistic Regression	50.00
Code 3	TruthfulQA	Random Forest	49.16
Code 3	TruthfulQA	XGBoost	49.61

B. English vs Arabic Comparison

As shown in Table IX and Figure 11, three key observations emerge. First, models trained on HaluEval show a consistent cross-lingual performance drop of 9–12%, showing that HaluEval-specific features do not transfer well to Arabic. Second, models trained on FActScore demonstrate a near-zero performance gap between English and Arabic, suggesting that atomic fact-based features are more language-neutral. Third, Logistic Regression trained on TruthfulQA achives surprisingly better result by 12.52% in English compared to Arabic, indicating structural similarities between TruthfulQA and AraHalluEval.

TABLE IX
ENGLISH VS ARABIC ACCURACY RESULTS (%)

Code	Model	English	Arabic	Gap
Code 1 (HaluEval)	LR	55.47	42.93	-12.54
Code 1 (HaluEval)	RF	54.90	44.93	-9.97
Code 1 (HaluEval)	XGBoost	54.87	45.16	-9.71
Code 2 (FActScore)	LR	41.43	45.76	+4.33
Code 2 (FActScore)	RF	51.86	50.66	-1.20
Code 2 (FActScore)	XGBoost	49.31	50.24	+0.93
Code 3 (TruthfulQA)	LR	37.48	50.00	+12.52
Code 3 (TruthfulQA)	RF	48.72	49.16	+0.44
Code 3 (TruthfulQA)	XGBoost	49.27	49.61	+0.34

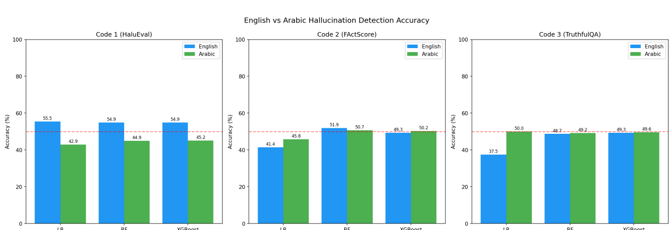


Fig. 11. Comparison between English and Arabic hallucination detection results.

C. Ablation Study

Table X and Figure 12 present the ablation study results for all three classical models tested on TruthfulQA. Across all models, removing Answer Length consistently improves accuracy with drops of -0.74%, -1.20%, and -1.27% for Logistic Regression, Random Forest, and XGBoost respectively confirming the presence of a strong Answer Length bias. Similarly, removing NLI Score has minimal or positive impact across all models, suggesting it contributes little to cross-dataset generalization. Entity Overlap is the most useful feature, removing it leads to the largest accuracy drop in Random Forest (0.93%) and XGBoost (0.46%). These findings suggest that the framework relies heavily on answer length as an indirect indicator for hallucination detection rather than semantic content, which limits its cross-dataset generalization capability.

TABLE X
ABLATION STUDY RESULTS TESTED ON TRUTHFULQA

Removed Feature	LR Baseline	LR Drop	RF Baseline	RF Drop	XGBoost Baseline	XGBoost Drop
Semantic Similarity	55.47%	+0.25%	54.90%	+0.34%	54.87%	+0.03%
NLI Score	55.47%	-0.57%	54.90%	0.00%	54.87%	-0.15%
Entity Overlap	55.47%	-0.03%	54.90%	+0.93%	54.87%	+0.46%
Answer Length	55.47%	-0.74%	54.90%	-1.20%	54.87%	-1.27%

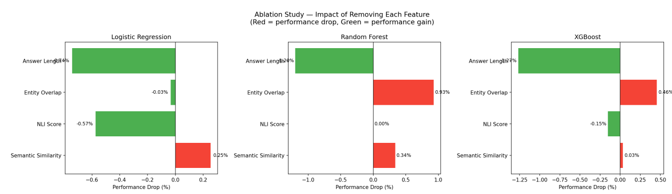


Fig. 12. Ablation study results across all models.

D. Efficiency Analysis

As shown in Table XI and Figure 13, all proposed models significantly outperform GPT-4 in terms of inference speed and model size. **Logistic Regression** achieves the fastest inference at 0.002 ms per sample, approximately 100,000 times faster than GPT-4. Among transformer models, **TinyBERT** offers the best efficiency balance, achieving competitive accuracy at 3.018 ms per sample with a model size of only 62.7 MB which is approximately 28,700 times smaller than GPT-4. These results confirm that the proposed framework provides a practical alternative to large language model-based hallucination evaluators.

TABLE XI
EFFICIENCY COMPARISON

Model	Speed (ms/sample)	Size
Logistic Regression	0.002	∼1 KB
XGBoost	0.008	∼5 MB
TinyBERT	3.018	62.7 MB
MiniLM	9.349	133 MB
DistilBERT	14.216	268 MB
GPT-4 (reference)	∼200	∼1,800 GB

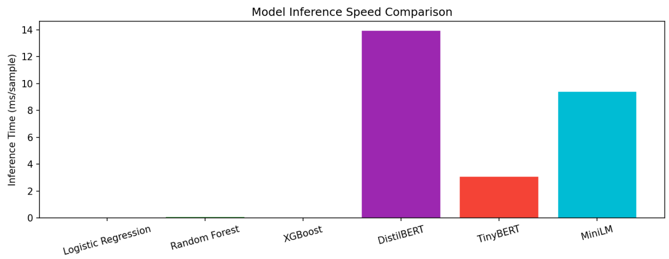


Fig. 13. Efficiency comparison of the evaluated models in terms of speed, size, and cost.

E. Explainability Analysis (SHAP)

As shown in Table XII and Figures 14–15, feature importance varies across training datasets. Answer Length dominates in Code 1 (SHAP=3.55), Semantic Similarity in Code 2 (0.68), and Entity Overlap in Code 3 (0.76), while NLI Score consistently shows the lowest importance across all codes.

TABLE XII
MEAN SHAP VALUES ACROSS TRAINING SETTINGS

Feature	**Code 1 → TQA**	**Code 2 → HE**	**Code 3 → FS**
Answer Length	3.55	0.47	0.44
Entity Overlap	2.05	0.32	0.76
Semantic Similarity	1.56	0.68	0.54
NLI Score	0.47	0.52	0.34

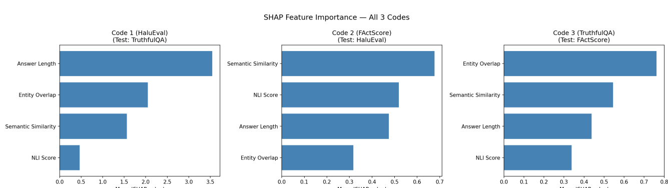


Fig. 14. SHAP feature importance results across all training settings.

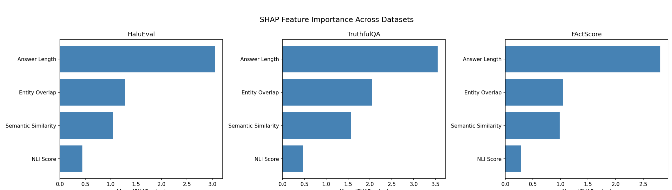


Fig. 15. SHAP-based feature importance comparison across datasets.

F. Error Analysis

As shown in Table XIII, the error analysis on TruthfulQA shows that the models produce different false-positive and false-negative patterns. TinyBERT detects many hallucinated samples but produces very few true negatives, while Logistic Regression provides the best overall accuracy in this setting.

TABLE XIII
ERROR ANALYSIS RESULTS ON TRUTHFULQA FOR CODE 1

Model	**TP**	**TN**	**FP**	**FN**	**Accuracy**
Logistic Regression	2,754	529	2,071	564	55.47%
Random Forest	2,762	487	2,113	556	54.90%
XGBoost	2,768	479	2,121	550	54.87%
DistilBERT	3,043	228	2,372	275	44.73%
TinyBERT	3,296	10	2,590	22	44.14%
MiniLM	2,091	1,227	1,882	718	52.53%

As shown in Figure 16, the error distribution on HaluEval reflects the difficulty of transferring models trained on other datasets to a nearly balanced benchmark.

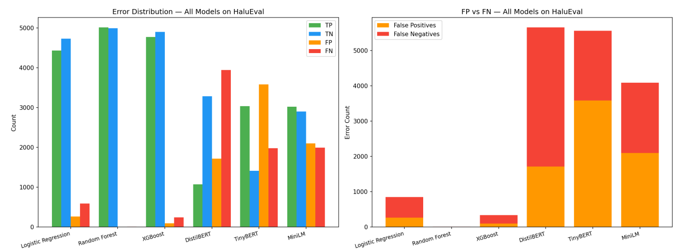


Fig. 16. Error analysis on HaluEval.

As shown in Figure 17, FActScore shows a different error profile due to its atomic-fact annotation and stronger class imbalance.

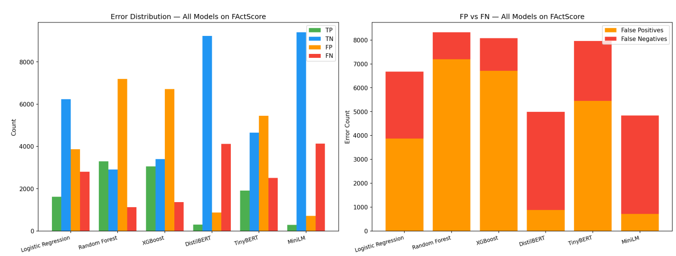


Fig. 17. Error analysis on FActScore.

As shown in Figure 18, TruthfulQA produces different error behavior because it focuses on truthfulness rather than direct factual support.

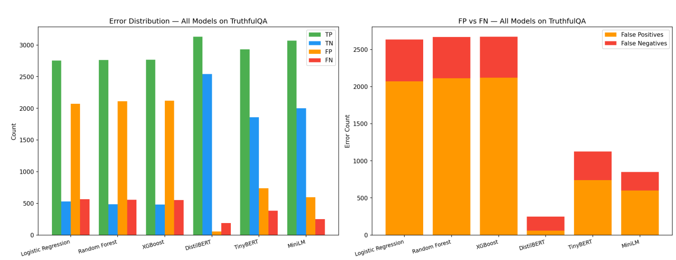


Fig. 18. Error analysis on TruthfulQA.

G. Baseline Comparison

As shown in Table XIV, the trained models generally outperform the random classifier and, in several cases, also outperform the majority-class classifier. MiniLM achieved the strongest result on FActScore, while Random Forest achieved the strongest result on HaluEval.

TABLE XIV
BASELINE COMPARISON RESULTS

Model	TruthfulQA	FActScore	HaluEval
Random Classifier	49.26%	49.21%	49.57%
Majority Classifier	56.07%	30.42%	50.10%
Best Code 1 (LR)	55.47%	54.05%	N/A
Best Code 2 (RF)	51.86%	N/A	58.44%
Best Code 3 (MiniLM)	N/A	66.66%	59.14%

As shown in Figure 19, the proposed models outperform simple baseline classifiers in several cross-dataset settings. MiniLM achieves the strongest result on FActScore, while Random Forest achieves the strongest result on HaluEval.

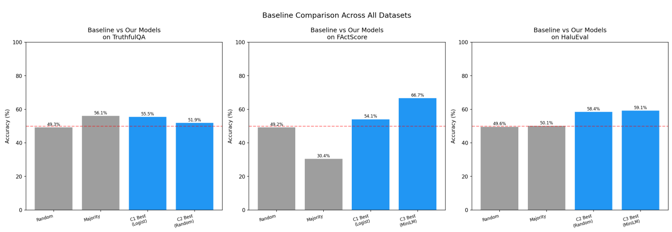


Fig. 19. Comparison between the proposed models and baseline classifiers.

H. Temperature Effect

To demonstrate the real-world utility of the proposed framework, a temperature effect experiment is conducted using Phi-2 [34], a 2.7 billion parameter language model developed by Microsoft. As shown in Table XV and Figure 20, a clear positive correlation is observed between generation temperature and hallucination rate. At temperature 0.1, only 20% of responses are detected as hallucinated, rising sharply to 80% at temperature 1.5 a fourfold increase. The hallucination rate remains relatively stable at low temperatures (0.1–0.9) before increasing significantly at temperatures above 1.0. These results confirm that higher generation temperature introduces greater randomness in LLM outputs, leading to increased hallucination, and demonstrate the ability of the proposed framework to detect such patterns in real LLM-generated responses.

TABLE XV
TEMPERATURE VS HALLUCINATION RATE

Temperature	Hallucination Rate (%)
0.1	20.0
0.3	30.0
0.5	20.0
0.7	30.0
0.9	30.0
1.1	50.0
1.3	70.0
1.5	80.0

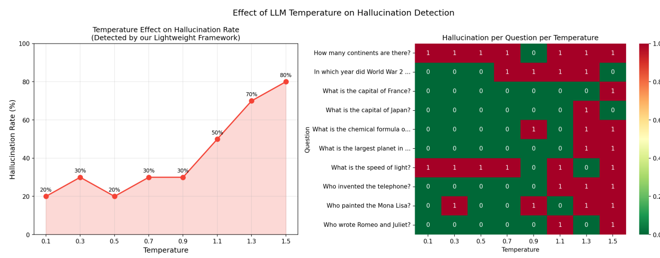
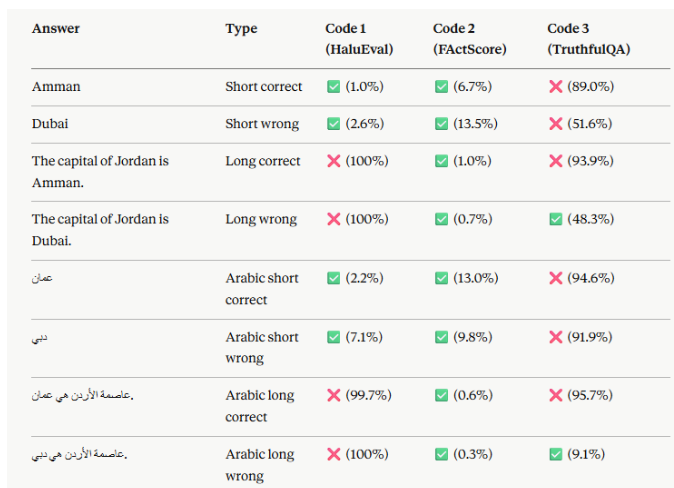


Fig. 20. Effect of temperature on detected hallucination rate.

I. Real Outputs

This section shows the real Hallucinated or non-Hallucinated results as shown in Figure 21



Answer	Type	Code 1 (HaluEval)	Code 2 (FActScore)	Code 3 (TruthfulQA)
Amman	Short correct	✅ (1.0%)	✅ (6.7%)	❌ (89.0%)
Dubai	Short wrong	✅ (2.6%)	✅ (13.5%)	❌ (51.6%)
The capital of Jordan is Amman.	Long correct	❌ (100%)	✅ (1.0%)	❌ (93.9%)
The capital of Jordan is Dubai.	Long wrong	❌ (100%)	✅ (0.7%)	✅ (48.3%)
عمان	Arabic short correct	✅ (2.2%)	✅ (13.0%)	❌ (94.6%)
دبي	Arabic short wrong	✅ (7.1%)	✅ (9.8%)	❌ (91.9%)
عاصمة الأردن هي عمان.	Arabic long correct	❌ (99.7%)	✅ (0.6%)	❌ (95.7%)
عاصمة الأردن هي دبي.	Arabic long wrong	❌ (100%)	✅ (0.3%)	✅ (9.1%)

Fig. 21. Manual test examples across the three training settings.

VI. CONCLUSION

This study presented a lightweight hallucination detection framework for identifying non-hallucinated and hallucinated answers generated by large language models. The proposed framework relies on a compact set of interpretable features, including semantic similarity, NLI score, entity overlap, and answer length. These features were used to train and evaluate six models: Logistic Regression, Random Forest, XGBoost, DistilBERT, TinyBERT, and MiniLM.

The framework was evaluated using three English hallucination benchmarks: HaluEval, TruthfulQA , and FActScore. To examine cross-dataset generalization, the experiments were organized into three experimental configuration, Where models were trained on one dataset and tested on the remaining unseen datasets. The best cross-dataset result was achieved by MiniLM with 66.66% accuracy when trained on TruthfulQA and tested on FActScore, while Random Forest achieved 58.44% when trained on FActScore and tested on HaluEval.

The ablation study confirmed that Answer Length is the dominant feature with a mean SHAP value of 3.55, introducing a systematic bias that limits cross-dataset generalization. The efficiency analysis demonstrated that Logistic Regression achieves inference speeds up to 100,000 times faster than GPT-4 at zero cost, while TinyBERT offers the best efficiency trade-off among transformer models at 3.018 ms per sample and 62.7 MB model size.

An Arabic extension was conducted using AraHalluEval, achieving a best accuracy of 50.66% using Random Forest trained on FActScore. Furthermore, a temperature effect experiment using Phi-2 demonstrated that higher generation temperature significantly increases hallucination rates, rising from 20% at temperature 0.1 to 80% at temperature 1.5. Overall, the proposed framework provides an efficient and interpretable approach for hallucination detection across multiple datasets and languages.

VII. Limitations and Future Work

A. Limitations

Despite the promising results of the proposed framework, several limitations are acknowledged:

1) Due to GPU memory constraints, transformer-based models were evaluated on English datasets only. The Arabic evaluation was limited to classical models, as loading transformer models alongside Phi-2 exceeded the available VRAM of 15.6GB.
2) Random Forest exhibited overfitting during training in two of the three experimental codes, achieving near-perfect training accuracy of 99.99% and 99.94% on HaluEval and FActScore respectively. Although regularization was applied in Code 3, a consistent strategy was not applied across all codes.
3) The Answer Length feature was identified as the dominant predictor (SHAP=3.55), introducing a systematic bias where short answers are consistently classified as factual and long answers as hallucinated, regardless of actual correctness.
4) Wikipedia-based retrieval support was initially planned as a fifth feature but was excluded due to unreliable entity extraction and high latency.

B. Future Work

The following directions are identified for future work:

1) Evaluating transformer models on Arabic hallucination data using multilingual models such as mBERT or AraBERT.
2) Replacing the Answer Length feature with more semantically meaningful features to reduce bias and improve cross-dataset generalization.
3) Incorporating Wikipedia or external knowledge base retrieval as a verification feature.
4) Extending the framework to other low-resource languages beyond Arabic to further validate cross-lingual transferability.

References

[1] Ji, Ziwei, et al. ”Survey of hallucination in natural language generation.” ACM computing surveys 55.12 (2023): 1-38.

[2] Kong, Linggang, et al. ”Multi-perspective consistency checking for large language model hallucination detection: a black-box zero-resource approach.” Frontiers of Information Technology & Electronic Engineering 26.11 (2025): 2298-2309.

[3] Woesle, Christian, Leopold Fischer-Brandies, and Ricardo Buettner. ”A Systematic Literature Review of Hallucinations in Large Language Models.” IEEE Access (2025).

[4] Yang, Borui, et al. ”Hallucination detection in large language models with metamorphic relations.” Proceedings of the ACM on Software Engineering 2.FSE (2025): 425-445.

[5] Wang, Cunxiang, et al. ”Survey on factuality in large language models.” ACM Computing Surveys 58.1 (2025): 1-37.

[6] Huang, Lei, et al. ”A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions.” ACM Transactions on Information Systems 43.2 (2025): 1-55.

[7] Reddy, G. Pradeep, YV Pavan Kumar, and K. Purna Prakash. ”Hallucinations in large language models (LLMs).” 2024 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream). IEEE, 2024.

[8] Yang, Bo, et al. ”Advancing LLM-Generated Code Reliability: A Hybrid Approach for Hallucination Detection.” IEEE Transactions on Software Engineering (2025).

[9] Barathidass, Charles, and Karthik Namburu. ”Rapid End-to-End Test Generation and Hallucination Mitigation using Generative Artificial Intelligence.” IEEE Access (2026).

[10] Ajmal, Rana Hassan, et al. ”Evaluating the Effectiveness of Advanced Language Models in Detecting and Mitigating Hallucinations Using Structured Question-Answering, Novel Metrics, and Post-Hoc Retrieval.” IEEE Access (2025).

[11] Al Umri, Najia, Andrea Stevens Karnyoto, and Bens Pardamean. ”The Impact of AI-Generated Hallucinations in Educational Settings: Trends, Gaps, and Future Directions.” 2025 9th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE). IEEE, 2025.

[12] Rebuffel, Clément, et al. ”Controlling hallucinations at word level in data-to-text generation.” Data Mining and Knowledge Discovery 36.1 (2022): 318-354.

[13] Wei, Zizhong, et al. ”Detecting and mitigating the ungrounded hallucinations in text generation by llms.” Proceedings of the 2023 International Conference on Artificial Intelligence, Systems and Network Security. 2023.

[14] Chen, Zhiyuan, et al. ”A survey of multimodal hallucination evaluation and detection.” International Journal of Computer Vision 134.3 (2026): 131.

[15] Malin, Ben, Tatiana Kalganova, and Nikolaos Boulgouris. ”A review of faithfulness metrics for hallucination assessment in large language models.” IEEE Journal of Selected Topics in Signal Processing (2025).

[16] Lin, Shanshan, et al. ”Constrained Paraphrase Consistency for LLM Hallucination Detection.” ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2026.

[17] Athreya, Deeksha, and Animesh Giri. ”From Randomness to Predictability-Fine Tuning the Key Parameters of LLM for Better Detection of Hallucination.” 2025 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT). IEEE, 2025.

[18] Koll, Thrilok, Tina Babu, and Rekha R. Nair. ”Evaluating Hallucinations in Large Language Models (LLM): Metrics and Mitigation.” 2025 IEEE 6th India Council International Subsections Conference (INDISCON). IEEE, 2025.

[19] Su, Weihang, et al. ”Mitigating entity-level hallucination in large language models.” Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region. 2024.
[20] Chen, Yuyan, et al. ”Hallucination detection: Robustly discerning reliable answers in large language models.” Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. 2023.
[21] Yadav, Sourabh, and Nishchal K. Verma. ”A Hybrid Framework for Hallucination Detection in Large Language Models.” IEEE Transactions on Artificial Intelligence (2026).
[22] Perković, Gabrijela, Antun Drobnjak, and Ivica Botički. ”Hallucinations in llms: Understanding and addressing challenges.” 2024 47th MIPRO ICT and electronics convention (MIPRO). IEEE, 2024.
[23] Wang, Zhenyu, et al. ”Large language modeling of hallucinatory problem mitigation based on the wheel of emotions.” Neural Networks (2025): 107996.
[24] Rahman, Subhey Sadi, et al. ”Hallucination to truth: a review of fact-checking and factuality evaluation in large language models.” Artificial Intelligence Review (2026).
[25] Liu, Yanyi, et al. ”Reducing hallucinations of large language models via hierarchical semantic piece.” Complex & Intelligent Systems 11.5 (2025): 231.
[26] Tikka, Petri, et al. ”Large Language Model hallucination mitigation in three industrial use cases.” IEEE Access (2026).
[27] Li, Junyi, et al. ”Halueval: A large-scale hallucination evaluation benchmark for large language models.” Proceedings of the 2023 conference on empirical methods in natural language processing. 2023.
[28] Lin, Stephanie, Jacob Hilton, and Owain Evans. ”Truthfulqa: Measuring how models mimic human falsehoods.” Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers). 2022.
[29] Min, Sewon, et al. ”Factscore: Fine-grained atomic evaluation of factual precision in long form text generation.” Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. 2023.
[30] Alansari, Aisha, and Hamzah Luqman. ”Arahallueval: A fine-grained hallucination evaluation framework for arabic llms.” Proceedings of The Third Arabic Natural Language Processing Conference. 2025.
[31] Sanh, Victor, et al. ”DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter.” arXiv preprint arXiv:1910.01108 (2019).
[32] Jiao, Xiaoqi, et al. ”Tinybert: Distilling bert for natural language understanding.” Findings of the association for computational linguistics: EMNLP 2020. 2020.
[33] Wang, Wenhui, et al. ”Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers.” Advances in neural information processing systems 33 (2020): 5776-5788.
[34] Javaheripi, Mojan, et al. ”Phi-2: The surprising power of small language models.” Microsoft Research Blog 1.3 (2023): 3.